NEURALRETAIL

AI-Powered Sales Intelligence & Predictive Analytics Platform

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INTERNSHIP PROJECT REPORT

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Organization

Amdox Technologies

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Acknowledgement

We sincerely thank Amdox Technologies for the opportunity to work on this internship project and gain practical exposure in Business Intelligence and Data Analytics.

Our heartfelt gratitude goes to our mentor, Mr. Chandan Mishra, for his continuous guidance, technical support, and valuable suggestions throughout the project. His expertise helped us gain deeper insights into retail analytics, dashboard development, inventory optimization, and predictive analytics.

We also thank everyone who directly or indirectly contributed to the successful completion of this project. The internship significantly enhanced our technical skills and understanding of real-world business intelligence solutions.

Abstract

Retail businesses generate enormous volumes of sales, customer, inventory, and transaction data every day. Transforming this data into meaningful insights is essential for improving operational efficiency, profitability, customer satisfaction, and inventory management.

NeuralRetail is an AI-Powered Sales Intelligence & Predictive Analytics Platform developed during the internship at Amdox Technologies. The platform integrates advanced Tableau dashboards with forecasting techniques to deliver a comprehensive business intelligence solution for retail organizations.

The platform consists of thirteen interactive dashboards covering executive reporting, sales performance, customer analytics, product analysis, regional insights, profitability monitoring, inventory risk management, customer churn intelligence, and simulation-based business analysis.

The project demonstrates how Business Intelligence and Predictive Analytics can transform raw retail data into actionable insights that support strategic planning and business growth.

Chapter 1: Introduction

1.1 Introduction

In today's competitive retail environment, businesses generate large volumes of transactional, customer, inventory, and operational data. Analyzing this data effectively is critical for improving profitability, customer satisfaction, inventory management, and overall business performance.

NeuralRetail was developed as an integrated analytics platform combining sales intelligence, customer analytics, inventory optimization, and predictive analytics into a unified decision-support system, utilizing Tableau dashboards to provide stakeholders with a comprehensive view of business operations.

1.2 Problem Statement

- Difficulty monitoring sales performance across products, customers, and regions
- Limited understanding of customer purchasing behavior
- Challenges in identifying profitable and underperforming products
- Inventory stockouts and excess inventory accumulation
- Lack of forecasting capabilities for future planning
- Difficulty identifying customer retention risks

1.3 Objectives & Scope

- Analyze retail sales performance and monitor profitability trends
- Evaluate customer behavior and segmentation
- Analyze product performance and revenue contribution
- Monitor inventory health and identify stock-related risks
- Evaluate regional business performance
- Support forecasting and predictive decision-making
- Develop an interactive Business Intelligence platform using Tableau

Chapter 2: Technology Stack

Technology	Purpose
Tableau Public	Primary BI tool for interactive dashboards, KPI monitoring, trend analysis, and forecasting visualizations
Python	Data preprocessing, analytical calculations, and forecasting operations
Pandas & NumPy	Data cleaning, manipulation, aggregation, and numerical analysis
Microsoft Excel	Preliminary data validation, inspection, and business metric verification
CSV Dataset	Retail sales dataset containing customer, product, sales, inventory, and payment data

Chapter 3: Dataset Description

The dataset consists of retail transaction records supporting sales analysis, customer segmentation, product performance evaluation, regional analysis, inventory optimization, and forecasting.

Data Group	Attributes
Customer Information	Customer Name, Segment, Region, Country
Product Information	Product Name, Category, Sub-Category
Sales Information	Order ID, Date, Quantity, Revenue, Profit, Discount, Shipping Cost
Inventory Information	Stock Quantity, Inventory Risk Status, Reorder Status, Dead Stock Flag, Critical SKU
Payment Information	Credit Card, Debit Card, PayPal, Cash
Forecasting Attributes	Forecast Value, LSTM Prediction, Risk Level Classification

Chapter 4 & 5: Methodology & Data Preparation

4.1 Development Workflow

- Data Collection – Retail transaction data collected and imported into the analytical environment
- Data Cleaning – Removal of duplicates, handling missing values, standardization of formats
- Feature Engineering – Creation of business metrics: Customer Revenue, Inventory Risk Status, Forecast Metrics, ABC-XYZ Classifications
- Dashboard Development – KPI identification, chart selection, layout design, filter integration, and testing in Tableau Public
- Insight Generation – Extraction of meaningful business insights supporting sales optimization, customer retention, and inventory planning

5.1 Feature Engineering Highlights

- Date Transformations: Year, Quarter, Month, Weekday for time-based trend analysis
- Customer Features: Total Customer Revenue, Customer Contribution, Repeat Purchase Analysis
- Product Features: Product Revenue, Profit, Ranking, Category Performance
- Inventory Features: Critical SKU Count, Dead Stock ID, Stockout Risk, Reorder Indicator, ABC-XYZ Classification
- Forecast Features: Forecast Value, LSTM Prediction, Risk Classification

Chapter 6 & 7: Dashboard Development & Analysis

A total of thirteen interactive dashboards were developed using Tableau Public, designed around simplicity, KPI-driven reporting, consistent visual design, and interactive navigation.

Dashboard 2 – Executive Summary

KPIs: Total Sales ($0.48M), Total Profit ($0.16M), Total Orders (2,000), Quantity Sold (7,115), Customers (1,534), Profit Margin (32.79%)

Figure 2: Executive Summary Dashboard

Key Findings: Sales stable throughout reporting period. Furniture generated highest revenue. Profit margins healthy at 32.79%.

Dashboard 3 – Sales Overview

Analyzes revenue growth and sales trends with 3-Year Revenue Trend, Month-over-Month Growth, and Quarterly Performance (2023–25).

Figure 3: Sales Overview Dashboard

Key Findings: Revenue increased steadily. Q4 generated strongest performance. October was the highest-performing month.

Dashboard 4 – Product Performance

Evaluates product contribution to revenue ($485K) and profitability ($159K) using Revenue vs Profit Scatter, Category Treemap, Top/Bottom 10 Products, and ABC Classification.

Figure 4: Product Performance Dashboard

Key Findings: Furniture ($256K) led revenue. Technology second ($140K). Small group drove majority of profits.

Dashboard 5 – Customer Analytics

Analyzes 1,534 customers — avg lifetime value $316, repeat purchase rate 25.6%. Includes Customer by Segment, Revenue Share, Top 10 Customers, and Spend Distribution.

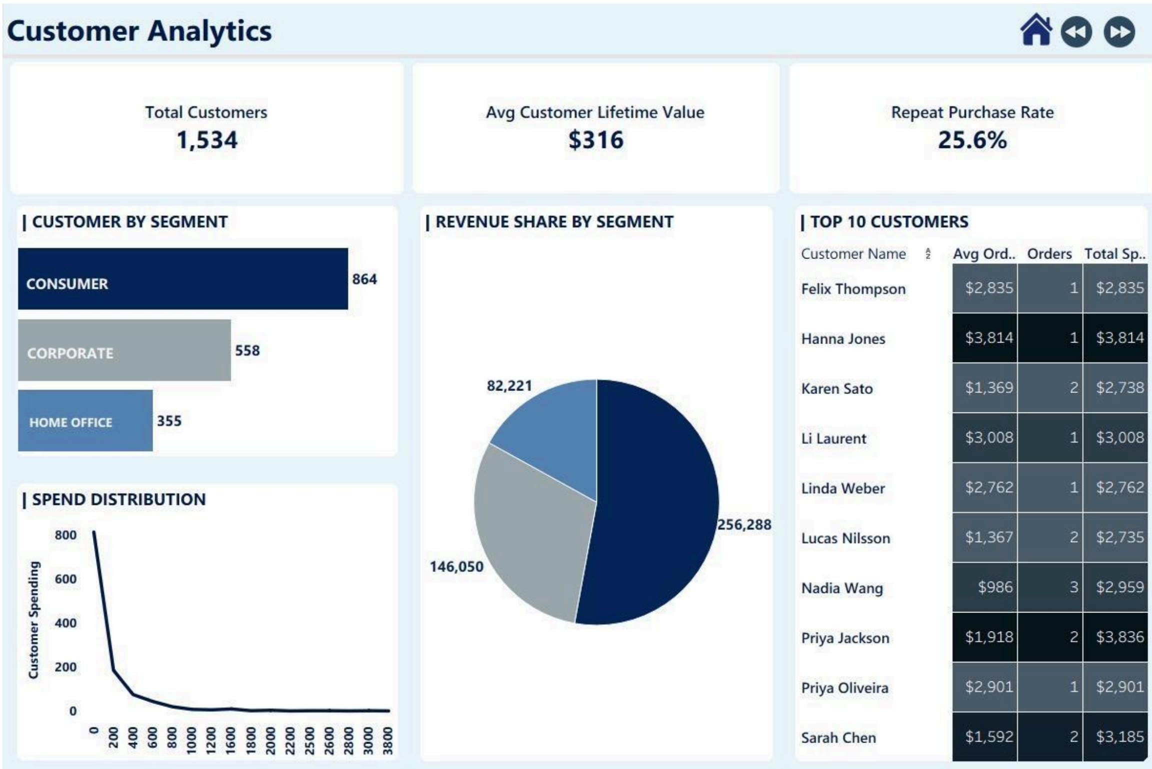

Figure 5: Customer Analytics Dashboard

Key Findings: Consumer segment largest (864). High-value customers contributed large share of total sales.

Dashboard 6 – Regional Insights

Evaluates geographic performance via Country Sales and Region Sales charts.

Figure 6: Regional Insights Dashboard

Key Findings: Europe top region ($137,006). North America second ($133,876). Mexico top country ($47,217).

Dashboard 7 – Monthly Insights

Monthly Sales Trend, Monthly Profit Trend, and Seasonal Sales Analysis for pattern identification.

Figure 7: Monthly Insights Dashboard

Key Findings: October highest sales ($51.1K). Q4 maximum revenue ($136,145). Q1 lowest ($102,251).

Dashboard 8 – Customer Revenue

Sales & Profit by Segment, Top Customers by Revenue, and Revenue by Payment Method breakdown.

Figure 8: Customer Revenue Dashboard

Key Findings: Consumer segment generated highest revenue. Credit Card dominant ($212,342). PayPal second ($156,171).

Dashboard 9 – Purchase Insights

2,000 orders — avg order value $242.3, avg 3.6 items/order, avg discount 8.6%. Includes Monthly Order Volume, Payment Methods, Orders by Day, and Top 5 Customer Transactions.

Figure 9: Purchase Insights Dashboard

Key Findings: Sunday highest order volume (303). Credit Card and PayPal preferred. Most orders in low-to-medium value range.

Dashboard 10 – Profit Trend

Revenue by Country, Profit by Region, Monthly Profit Trend, and Revenue vs Profit Scatter Plot.

Figure 10: Profit Trend Dashboard

Key Findings: Europe led profit by region ($45,672). Revenue and profit maintained strong positive relationship throughout.

Dashboard 11 – Inventory Risk

Stockout Risk Table, Risk Distribution, ABC-XYZ Matrix, and Dead Stock Items analysis.

Critical SKUs	Dead Stock Items	Stockout Rate	Reorder Needed
3	21	7.50%	35 products

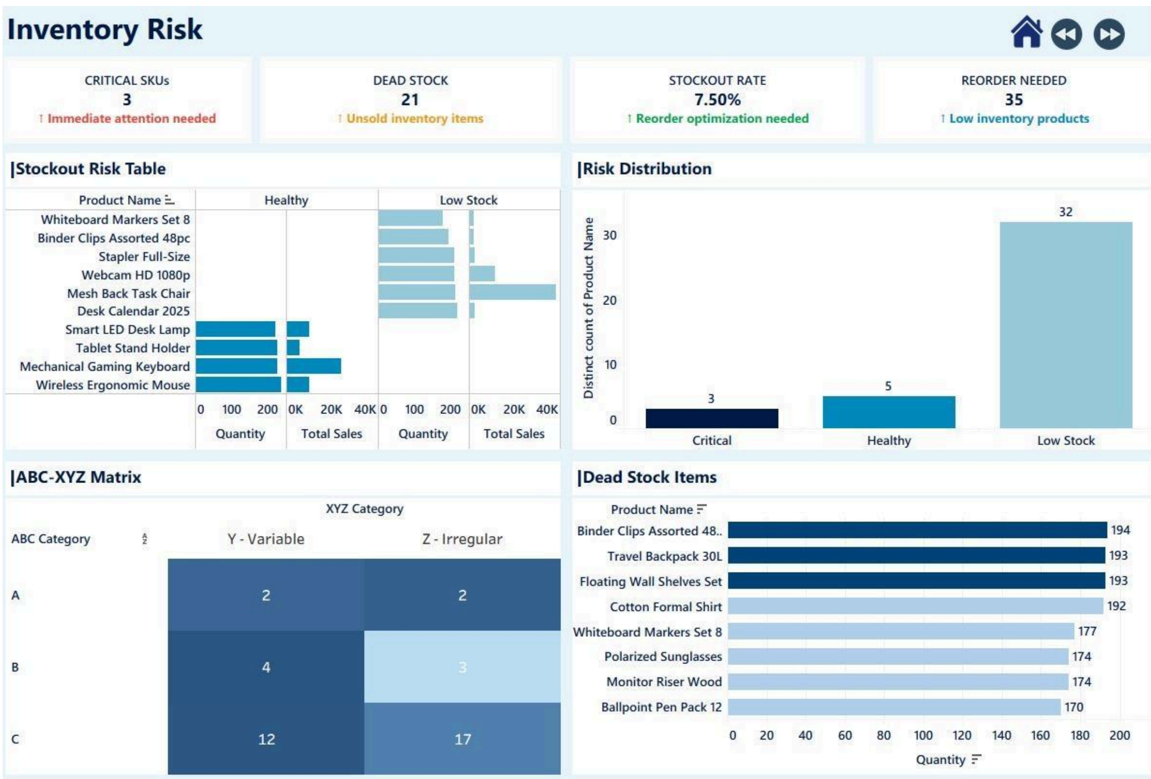

Figure 11: Inventory Risk Dashboard

Key Findings: 21 dead stock products identified. 3 critical SKUs needed immediate attention. Low Stock (32) was largest category.

Dashboard 12 – Churn Intelligence

AUC-ROC 1.00 | MAPE 9.4% | LSTM Prediction 20,391 | Forecast 10.70 | High Risk Customers: 1,413

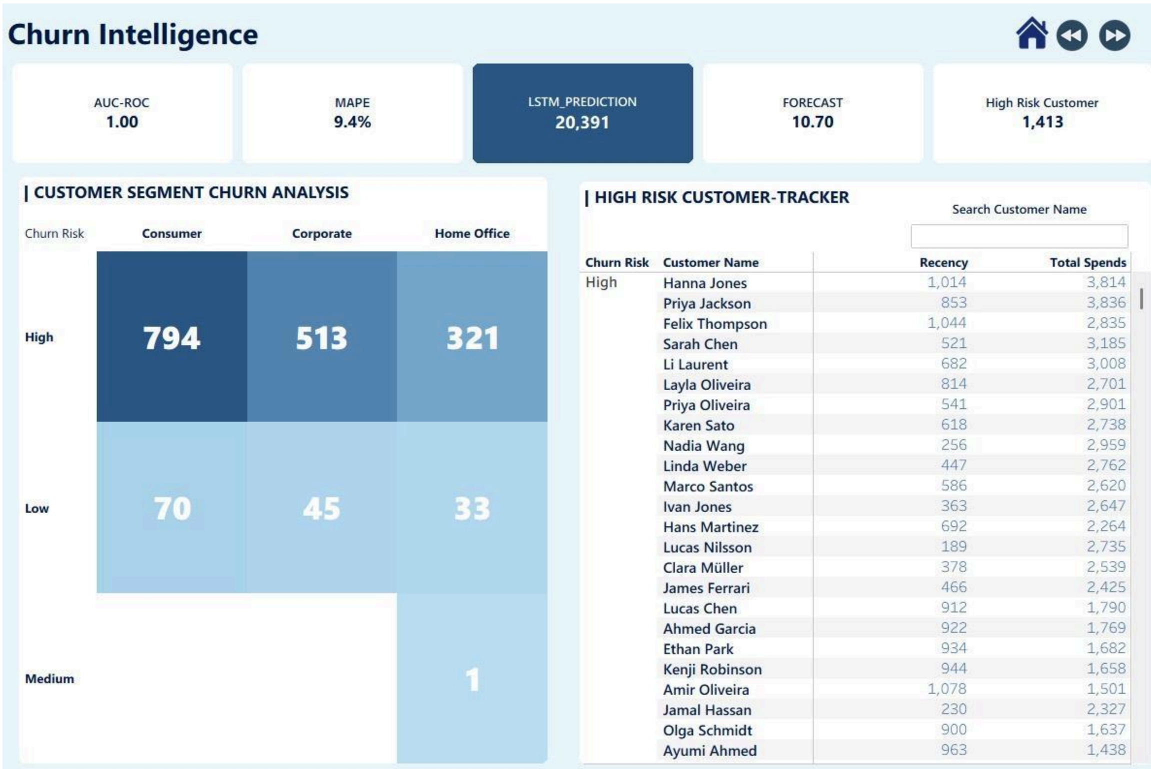

Figure 12: Churn Intelligence Dashboard

Key Findings: Consumer had largest high-risk group (794). Corporate 513, Home Office 321 high-risk customers.

Dashboard 13 – Filters & Simulation

Year, Region, Product Category, and Customer Segment filters with a What-If Simulator for real-time discount impact analysis.

Figure 13: Filters & Simulation Dashboard

Enables self-service analytics and dynamic scenario exploration without modifying the underlying dataset.

Chapter 8: Predictive Analytics & Forecasting

Forecasting Area	Description & Benefits
Sales Forecasting	Estimates future revenue trends; supports demand planning and resource allocation
Customer Risk Forecasting	Identifies customers needing proactive retention; improves segmentation and CRM
Inventory Forecasting	Stock planning via Forecast Value, LSTM Prediction, Risk Classification; reduces stockout risk

Chapter 9: Key Business Insights

9.1 Sales Insights

- October was the highest-performing month
- Quarter 4 generated the strongest sales performance
- Revenue remained relatively stable throughout the reporting period
- Seasonal demand significantly influenced sales trends

9.2 Customer Insights

- Consumer customers represented the largest customer segment
- Repeat customers contributed significantly to overall revenue
- High-value customers generated a large share of business profits

9.3 Product Insights

- Furniture was the highest-performing product category
- A small product group drove the majority of profits
- Some products delivered high sales but lower profitability

9.4 Regional Insights

- Europe emerged as the strongest-performing region
- Mexico generated significant sales revenue as the top country
- Regional sales performance varied considerably across markets

9.5 Inventory Insights

- 21 products identified as dead stock requiring immediate action
- Low Stock products represented the largest inventory category
- Inventory risk monitoring improved overall stock visibility

9.6 Forecasting Insights

- Forecast outputs supported demand estimation and future planning
- Risk classifications improved inventory planning decisions
- Customer risk indicators supported proactive retention planning

Chapter 10: Results & Outcomes

Outcome	Description
Interactive Dashboard Ecosystem	13 interactive dashboards providing complete analytical view of retail operations
Centralized Business Intelligence	Sales, customer, product, inventory, and forecasting analytics unified in one platform
Improved Decision-Making	Raw retail data transformed into actionable insights for strategic decisions
Enhanced Inventory Visibility	Identification of stockout risks, dead stock products, and replenishment requirements
Better Customer Understanding	Insights into purchasing behavior, customer segments, and revenue contribution
Forecasting Capabilities	Forecasting indicators and risk assessments for planning and inventory optimization

Chapter 11: Challenges Faced

11.1 Data Preparation Challenges

- Data Quality: Dataset contained inconsistencies requiring cleaning and validation
- Missing Values: Several fields required preprocessing to ensure analytical accuracy
- Standardization: Different formats needed unification before dashboard development

11.2 Dashboard Development Challenges

- Layout Design: Required multiple iterations to achieve usability and visual appeal
- KPI Selection: Selecting meaningful KPIs while maintaining simplicity was challenging
- Navigation Design: Building smooth inter-dashboard navigation required careful planning
- Visual Consistency: Maintaining uniform design style across all 13 dashboards

11.3 Analytical Challenges

- Inventory Classification: ABC-XYZ classifications required extensive testing
- Forecast Integration: Embedding forecasting outputs into Tableau required customization
- Insight Generation: Converting outputs into actionable recommendations required careful interpretation

All challenges were resolved through iterative development, continuous testing, and visualization best practices.

Chapter 12: Future Enhancements

Enhancement	Description
Real-Time Data Integration	Integrate live databases for real-time monitoring and reporting
Cloud Deployment	Deploy on cloud infrastructure for improved accessibility and scalability
Automated Alerts	Automated notifications for inventory shortages, critical SKUs, and customer risk
Enhanced Forecasting	Advanced forecasting techniques for improved prediction accuracy
Mobile Dashboard Support	Mobile-responsive dashboards for anywhere access
Multi-Store Analytics	Support for multiple retail locations and enterprise-wide reporting
AI-Powered Recommendations	Product recommendations, inventory optimization, and customer engagement suggestions

Chapter 13: Conclusion

NeuralRetail successfully demonstrates the integration of Business Intelligence, Retail Analytics, Inventory Optimization, and Predictive Analytics into a unified analytical ecosystem.

The project transformed retail transaction data into meaningful business insights through thirteen interactive Tableau dashboards, providing comprehensive visibility into sales performance, customer behavior, product profitability, regional performance, inventory risks, and forecasting indicators.

The platform successfully addressed key business challenges related to sales monitoring, customer analytics, inventory management, and strategic planning. Interactive visualizations enabled stakeholders to quickly understand business performance and make informed, data-driven decisions.

Overall, NeuralRetail demonstrates how Business Intelligence and Data Analytics can improve operational efficiency, strengthen decision-making, optimize inventory management, and drive business growth. The project provided valuable practical experience in Tableau dashboard development, retail analytics, KPI monitoring, forecasting, and business intelligence implementation.

Chapter 14: Project Contribution

The following table outlines the individual and joint contributions made by each team member throughout the NeuralRetail project.

Somyashree Nayak	Likhitha	Joint Contributions
• Cover Page Design	• Dashboard 6: Regional Insights	• Dataset Preparation
• Dashboard 1: Cover Page	• Dashboard 7: Monthly Insights	• Feature Engineering
• Dashboard 2: Executive Summary	• Dashboard 8: Customer Revenue	• Forecasting Analysis
• Dashboard 3: Sales Overview	• Dashboard 9: Purchase Insights	• Business Insights
• Dashboard 4: Product Performance	• Dashboard 10: Profit Trend	• Final Testing & Validation
• Dashboard 5: Customer Analytics	• Dashboard 11: Inventory Risk	
• Dashboard 12: Churn Intelligence	• Analysis Documentation	
• Dashboard 13: Filters & Simulation	• Insight Generation	
• Overall Dashboard UI/UX Design	• Data Preparation	
• Theme Selection		
• Navigation Design		
• Report Design & Formatting		
• Demo Video		

— End of Report —